

COLUMN | RAMAN SHARMA

2041: The Next 20 Years of Cloud Computing



Raman Sharma,
Vice President, Product & Programs
Marketing at DigitalOcean

Undoubtedly, the cloud has been a democratizing force in making the consumption and production of software accessible to a far bigger population than ever before. The meaning of cloud itself has changed over the years and is hard to point out exactly when the cloud came to life, but we know that the cloud will evolve to include much more in the future. Over time, all aspects of the cloud - Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) - will flourish.

Below are a few trends that we see early signs of becoming prominent in the coming years.

SaaS takeover would be near complete.

Today we primarily focus on IaaS and PaaS, essentially the building

blocks that software builders utilize for deploying and operating software applications. SaaS applications like CRM, ERP, Online Collaboration, Finance, Marketing, HR, Analytics, etc., have boosted the productivity of the economy overall. That trend will only grow as more people build SaaS applications, and more software builders will be offering their products thanks to the presence of intuitive IaaS and PaaS offerings.

The definition of a “developer” will become more inclusive.

Today a developer is someone who writes code, builds features, etc., to deploy and operate software, and the skills required to do it is still pretty technical. But, a new wave of tools and technologies - like serverless and Low-Code / No-Code tools - is enabling more people to create new softwares. Designers, Data Scientists, and many more will be able to utilize these tools to build software of all kinds. Some of these would be for internal use within businesses, however, even more exciting would be the many more external-facing SaaS applications made possible because more people with ideas will have tools to build these apps.

More infrastructure, more platform, more serverless, more Low-Code/No-Code

Much was predicted about IaaS fading away once more modern ways of building apps came along. However, IaaS usage across all clouds continues to grow. In the near future there will be people exercising all of these technologies to a greater degree than what we see today.

Multi-Cloud will not be a special case.
Going forward, we predict that we

will see a much lower resistance to multi-cloud, especially among smaller teams and businesses. To aid with multi-cloud adoption, newer management planes and observability tools will be available to allow developers to manage and operate their multi-cloud investments and architectures. Also, higher-level abstractions will be used by people to build applications - like PaaS, Serverless, and LowCode/NoCode tools - that will themselves be multi-cloud.

Cloud-native will pass lift-and-shift

A vast majority of computing still happens on on-premises data centres. Many of these applications are not utilizing the benefits of the public cloud fabric. In the future, we will see more applications getting built to take full advantage of the high availability, global reach, and economies of scale that the public cloud has to offer.

Some of this will be enabled by fully managed cloud services that are already built, but many other cloud-native applications will be built by teams adopting modern cloud architectures and tools like Kubernetes.

Blockchain and Machine Learning will no longer be just buzzwords.

Applications enabled by technologies like Blockchain and actual business use of Machine Learning are still pretty niche scenarios - limited only to people with expertise. But more people are beginning to utilize the advantages of the cloud to build solutions in these areas. Going forward, we will see growth in applications - like supply chain, healthcare, financial services, etc. - taking advantage of technologies powered by the cloud. Storage and PaaS are perfect candidates to take advantage of distributed architectures like Blockchain. Similarly, cloud management, like diagnostics, analytics, etc., can benefit from Machine Learning.

This list is by no means exhaustive. Our industry will invent new technologies in the next few decades but what will remain true is that the cloud will also power the future. 